

EXPERIENCE

Rural Studies Institute, GCSU (Sponsored by Georgia Tech PIN Program)

Milledgeville, GA

Front-End Developer & Data Visualization Intern

May 2025 – Aug 2025

- Designed and shipped interactive, data-driven maps for the Black Belt Regional Database using **JavaScript**, **Leaflet.js**, and WordPress; served **500+** monthly users.
- Built a front-end visualization layer that transformed complex socio-economic datasets into interactive, accessible maps for non-technical researchers and policymakers.
- Automated data ingestion pipelines converting spreadsheets to **GeoJSON** (10k+ data points), reducing integration time by **30%** and manual data entry by **40%**.
- Presented solutions at the *Emerging Innovators Celebration*.
- Project Link:** gcsu.edu/ruralstudies/understanding-black-belt-region

PROJECTS

TweenGen — Browser-Based Animation Editor | *React.js, Fabric.js, GSAP, Recoil, Vite* [GitHub](#) | [Demo](#)

- Architected a lightweight in-browser animation engine with custom easing and path morphing, reducing final bundle size by **15%** vs. external animation libraries.
- Achieved a consistent **60 FPS** during complex multi-object animations by cutting canvas re-render cycles by **40%** through memoization and selective Recoil state updates.
- Built a drag-and-drop keyframe timeline, GSAP-powered live preview, and code export pipeline generating production-ready HTML/CSS/JavaScript.
- Implemented bitmap-to-SVG vectorization, shape deformation, and audio waveform visualization using the Web Audio API.

Real-Time Collaborative Task Board | *React, TypeScript, Socket.io, Node.js, MongoDB* [GitHub](#) | [Demo](#)

- Built real-time collaborative drag-and-drop UI with optimistic updates, keeping **50+** concurrent users in sync at **<100ms** latency via WebSockets.
- Designed responsive interface with React Beautiful DnD, role-based views, and conflict resolution that reduced state sync errors by **15%** during peak activity.
- Deployed frontend on Vercel with automated CI/CD via GitHub Actions.

TECHNICAL SKILLS

- Core (Proficient):** JavaScript/TypeScript, React.js, HTML/CSS, Tailwind CSS, Recoil, Vite, Git
- Frontend Craft:** Next.js, Zustand, Fabric.js, GSAP, Canvas/SVG rendering, React Beautiful DnD, Leaflet.js
- Testing & Quality:** Jest, Vitest, Unit & Integration Testing
- Backend & Data:** Node.js, Express.js, RESTful APIs, MongoDB, PostgreSQL, Python
- DevOps & Tooling:** Vercel, GitHub Actions CI/CD, Docker, AWS (S3, Lambda, EC2), Postman, Zod

EDUCATION

The University of Georgia

Athens, GA

Master of Science in Computer Science

Aug 2024 – May 2026

- Coursework: Software Engineering, Database Management, Algorithms, Computer Networks, Data Science, Advanced Representation Learning

Sathyabama Institute of Science and Technology

Chennai, India

Bachelor of Engineering in Computer Science

Aug 2020 – May 2024

CERTIFICATIONS

- AWS Certified Developer – Associate ([link](#))
- AWS Certified Cloud Practitioner ([link](#))